

Distributed System

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Distributed System is a collection of autonomous computer systems that are physically separated but are connected by a centralized computer network that is equipped with distributed system software. The autonomous computers will communicate among each system by sharing resources and files and performing the tasks assigned to them.

Types of Distributed Systems

1. [Client/Server Systems](#)
2. [Peer-to-Peer Systems](#)
3. [Middleware](#)
4. [Three-tier](#)
5. N-tier

Client-server systems: the most traditional and simple type of distributed system, involve a multitude of networked computers that interact with a central server for data storage, processing or other common goal.

Peer-to-peer system: a computer network which enables peers to share the network resources, computational power and data storage, without relying on a central authority, distributing workloads among hundreds or thousands of computers all running the same software. The system functions as both a client and a server.

Characteristics of Distributed System

- **Resource Sharing:** It is the ability to use any Hardware, Software, or Data anywhere in the System.
- **Openness:** It is concerned with Extensions and improvements in the system (i.e., How openly the software is developed and shared with others)
- **Concurrency:** It is naturally present in Distributed Systems, that deal with the same activity or functionality that can be performed by separate users who are in remote locations. Every local system has its independent Operating Systems and Resources.
- **Scalability:** It increases the scale of the system as a number of processors communicate with more users by accommodating to improve the responsiveness of the system.
- **Fault tolerance:** It cares about the reliability of the system if there is a failure in Hardware or Software, the system continues to operate properly without degrading the performance of the system.
- **Transparency:** It hides the complexity of the Distributed Systems to the Users and Application programs as there should be privacy in every system.
- **Heterogeneity:** Networks, computer hardware, operating systems, programming languages, and developer implementations can all vary and differ among dispersed system components.

Challenges of Distributed Systems

- **Network latency:** The communication network in a distributed system can introduce latency, which can affect the performance of the system.
- **Distributed coordination:** Distributed systems require coordination among the nodes, which can be challenging because of the distributed nature of the system.
- **Data consistency:** Maintaining data consistency across multiple nodes in a distributed system can be challenging.

Advantages

1. High Performance
 - a. Distributed systems can outperform centralised systems by utilising the capabilities of several computers to handle the demand.
2. Reliable
 - a. In terms of failures, distributed systems are significantly more dependable than single systems. Even if a single node fails, it does not affect the remaining servers. Other nodes can continue to operate normally.
3. Scalable
4. Expandability
5. Availability
6. Reduced latency (short distance)
 - a. Low latency is achieved using distributed systems. If a node is closer to the user, the distributed system ensures that traffic from that node reaches the system. As a result, the user may realise that it takes significantly less time to serve them.
7. Secure
 - a. Distributed systems include security safeguards that prevent data breaches and illegal access to any data, hardware or software of an organization.
8. Cost-effectiveness
 - a. Distributed systems, despite their high expense of implementation, are reasonably cost-effective in the long term. In contrast to a mainframe computer, which is made up of multiple processors, a distributed system is made up of several computers working together. This infrastructure is significantly less expensive than a mainframe system.
9. Geographic Distribution
 - a. A feature of a distributed system is geographic distribution. It allows companies and organisations to provide users with services in different areas.

Disadvantages

1. High cost.
2. The problem is finding the fault.
3. More space is needed.
4. The increased infrastructure is needed.
5. In distributed systems, it is challenging to provide adequate security because both the nodes and the connections must be protected.